Time Series Wave Forecasting with Transformer Model, Case Study in Pelabuhan Ratu, Indonesia

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***Abstract*—Wave height prediction is vital in various fields of work and activities at sea, one of which is for ship navigation and operations. These activities depend on current ocean wave data and predictions for the future. This paper uses the Transformer model to predict ocean wave heights accurately. Although it is a new model, it effectively measures wave height. This study used six years and seven months of data divided into six years and seven months as training data and testing data, respectively. We used testing data to forecast the next three days, seven days, 14 days, and 30 days. As wave data, we use the reanalysis ERA5 data from ECMWF at Pelabuhan Ratu, Indonesia, obtained from ERA-5 as a case study. We compare the forecasting results to get the highest accuracy using Transformer and Long Short-Term Memory (LSTM). Our results show that the LSTM model has the best accuracy than the Transformer model. The highest accuracy of sea wave height forecasting for the LSTM model is 0.034 for MSE, 0.186 for RMSE, and 0.996 for R, and the best accuracy of sea wave height forecasting for the Transformer model is 0.118 for MSE, 0.186 for RMSE, and 0.992 for R.**

***Index Terms*—Wave height forecasting, Long Short-Term Memory (LSTM), Transformer.**

I. Introduction

Ocean waves are a natural phenomenon that often occurs in the sea. Analysis of wave height is necessary to ensure safety at sea or on the beach. The Pelabuhan Ratu Beach area is a place for tourism and sea transportation. Shipping and other maritime activities are highly dependent on wave conditions. Various techniques, model, and historical data predict future wave conditions [1]. High oceans waves can harm and damage ships and cause serious casualties. These can be prevented by optimizing the transportation route. As a result, predicting wave heights can increase productivity, save fuel, and avoid dangerous situations. [2]. Developing a system that can predict time series wave forecasting can help those active in the middle of the sea.

Predicting ocean wave heights is difficult due to their stochastic and non-linear properties [3]. Therefore, research on high prediction is carried out continuously to improve the prediction accuracy and computational speed of wave forecasting systems [4].

Several models have been created to predict wave height. Several models have been created to predict wave height. In 2019, Zhao [5] has researched wave height prediction using the Support Vector Machine (SVM) model and Empirical Mode Decomposition - Support Vector Machine (EMD-SVM) model. From the experimental results for 3,6,12, and 24 hours, it can be said that EMD-SVM is superior to SVM.

Recent approaches use machine learning, such as LSTM and Transformers.

This paper applies the Transformer model to predict ocean waves in Pelabuhan Ratu. It will be compared with the LSTM model to measure better performance.

The contents of this paper include the following sections. In Section 2 we describe a literature review on time series forecasting, the Transformer model, and the Long Short Term Memory model. Section 3 describes the research methodology and includes data explanation, data processing, and evaluation model, followed by Section 4 discusses the study results. Lastly, the paper is concluded in Section 5.

II. Literature Review

A. Time Series Wave Forecasting

Time Series is a time-focused field of machine learning. As the name suggests, this field focuses on analyzing the data sequence over time and predicting future data based on previous data.

Various researchers have carried out the development of a wave forecasting system, one of which is Zhang [4], who conducted a study to measure the estimated wave height in the Bohai Sea and Xiaomaidao using N-LSTM model. The observed consequences reveal that the N-LSTM network has better prediction accuracy. Moreover, the N-LSTM model performs better when the effective wave height is more than 1.25 m, and the numerical model predicts poor performance in this case. The N-LSTM model makes probabilistic predictions for significant wave heights, and the prediction confidence interval covers most of the real observations. Therefore, the N-LSTM model can overcome difficult wave situations and predictive uncertainty and create strong predictive strategies.

In 2021, Zhou [6] surveys were conducted using the convolutional LSTM model in the South China Sea and the East China Sea. Under normal conditions, for six, 12, and 24 hours, the correlation coefficients were 0.98, 0.93, and 0.83, and the average fundamental allocation error was 15%,

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29%, and 61%, respectively. Based on these findings, we can conclude that time-series ConvLSTM applies to 2D waves and produces highly efficient accuracy. This article proposes a machine learning technique for predicting wave height time series utilizing the Transformer and LSTM model.

B. Transformer

The Transformer model was first published in 2017 in Google's paper," Attention is all you need," which submitted a new technique named" self-attention." The Transformer model [7] originally proposed for machine translation, the Transformer model was quickly integrated into other areas such as image production [8], voice [9], text summarization [10], and music due to its high performance [11]. Transformer models concentrate on modeling, not iterations and convolutions. The transformer uses an encoder-decoder approach. First, the input is sent to the encoder. After encoding, the resulting output will be the same as the encoded input and the previous decoder output. Transformer architecturer [12] is shown in Figure 1.

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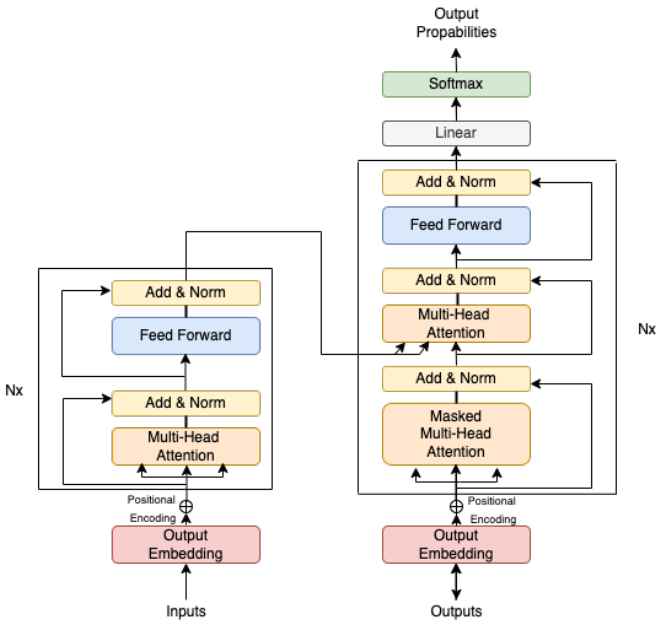


Figure 1. The architecture of Transformer [12]

The transformer uses stacked self-attention and pointwise layers that are fully connected to the encoders and decoders shown in Figure 1. The encoder layer maps every input sequence to a continuous abstract representation that stores the information learned throughout the sequence. Figure 1 shows the use of two submodules to show that a fully connected network follows multi-head attention. Also, there are remaining connections around each of the two sublayers, followed by the normalization layer.

C. Long Short Term Memory

Hochreiter and Schmidhuber first presented LSTM in 1997 [13]. LSTM is included in the RNN class, which is specifically capable of analyzing time series [14]. Various researchers, including in 2021, carried out the development of wave prediction systems using the LSTM model, Zhou [14], who measured estimated waves using the EMD-LSTM model using LSTM. EMD-LSTM is superior to LSTM, even when measured using predictive evaluation metrics (MSE, MPE, and R).

In current years, lengthy short-time period memory (LSTM) networks with apparent blessings have exhibited overall recordable version on plenty of missions, which include photograph viewing, language translation [15], and handwriting recognition.

In 2016, Rui F [16] used LSTM and neural network (NN) techniques for gated recurrent units (GRUs) to predict short-term traffic flows. In experiments, deep learning methods based on recurrent neural networks (RNNs) are superior to LSTMs and GRUs over autoregressive integrated moving average (ARIMA) model.

III. METHODOLOGY

In this analysis, the wave height was forecasted utilizing the transformer model. As a case analysis, we used wave data from Pelabuhan Ratu. Data was taken from ERA-5, totaling 66449 data from January 2014 to July 2021. The dataset was then labeled and split into training and testing data. First, we processed the plain wave data. We processed the dataset and separated it into training data and testing data in this phase. The previously processed data was used as input to the Transformer model. In this article, we used the Transformer model to forecast the wave height time series data and the previous time step as input to predict the next step. This is commonly referred to as the sliding window model. The methodology of this research is shown in Figure 2

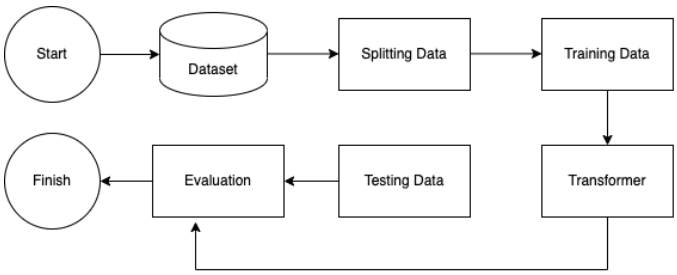


Figure 2. Methodology for Wave Height Forecasting.

A. Wave Data

This case uses the data in Pelabuhan Ratu, Indonesia, with the coordinates of 7°15'00.0" S 106°00'00.0" E. Figure 3

We used wave data from ERA-5 historical data, which had a temporal duration of 1 hour and spanned from January 1, 2014 until July 31, 2021. The data was divided into 80% training data and 20% testing data.

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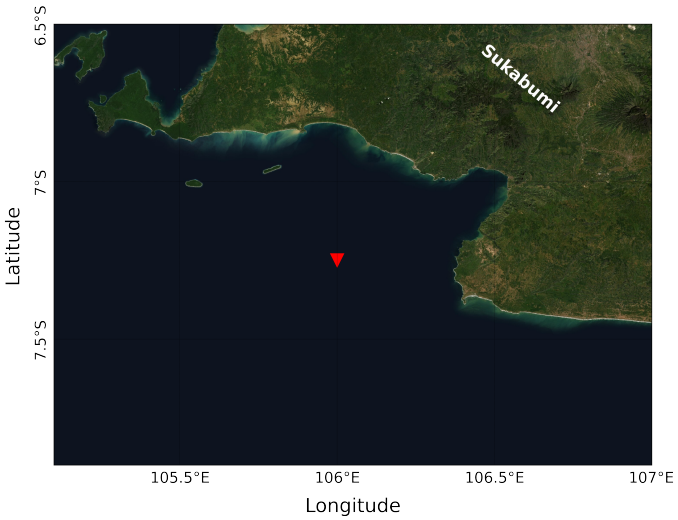


Figure 3. Location of Pelabuhan Ratu, Indonesia.

B. TSAI

The library used was from the Time Series Artificial Intelligence [tsai] with the TSForecaster model and the TSTPlus architecture [17].

C. Performance Analysis

We used Mean Square Error (MSE) [18], Root Mean Square Error (RMSE) [18], and Coefficient Correlation (R) [19] to research model performance. MSE is the sum of the squared contrast between factual significances and the expected/estimated steals, which can be formulated into the following equation:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 \quad (1)$$

where n representing the number of data, Y representing the actual value, and $\bar{y}$ representing the predicted value.

$$RMSE = \sqrt{MSE} \quad (2)$$

where RMSE representing the root of the MSE value.

$$R = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}i)}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \quad (3)$$

Where $\bar{x}$ representing the average value of the variable x, $\bar{y}$ representing the average value of the variable y, and n representing the number of data.

IV. Result and Discussion

In this section, we compared wave height forecasting results. Here, we define five scenarios of forecasting based on the length of days to be forecasted, i.e., 3, 7, 14, and 30 days ahead. We use six years data for training process of the machine learning models, as displayed in Figure 4. However, just one month's data was used for testing due to scenarios. The longest visualized forecast was 30 days. This was displayed to see the performance of the Transformer and LSTM model. Because 30 days is ideal for testing sequence data on transformer model and LSTM model

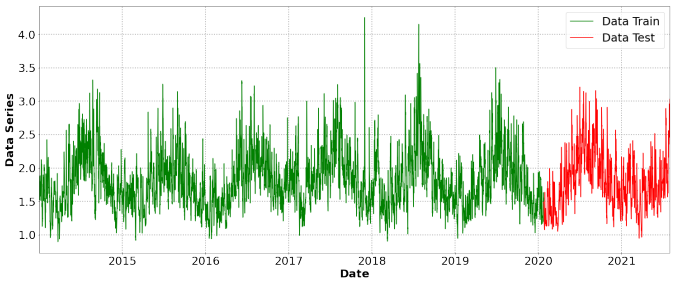


Figure 4. Training and Testing Data.

Table I shows results of prediction using the Transformer model. The three-day forecast has the worst value for MSE, RMSE, and R, i.e., 0.204 for MSE, 0.451 for RMSE, and 0.910 for R. On the other side, 30 days forecast gives the best MSE, RMSE, and R values, i.e., with 0.118 for MSE, 0.344 for RMSE, and 0.992 for R.

TABLE I
Results of prediction using Transformer model.

Prediction (Days)	MSE	RMSE	R
3 Days	0.204	0.451	0.910
7 Days	0.168	0.410	0.961
14 Days	0.125	0.354	0.992
30 Days	0.118	0.344	0.992

Table II shows results of prediction by using the LSTM model. The best MSE, RMSE, and R values is achieved for the prediction of 30 days, which are 0.034, 0.186, and 0.996, respectively. Predictions for 30 days have the best MSE, RMSE, and R values. As in the Transformer model, the three-day prediction has the lowest performance value of MSE, RMSE, and R, i.e., 0.052 for MSE, 0.229 for RMSE, 0.910 for R.

TABLE II
Results of prediction using LSTM model.

Prediction (Days)	MSE	RMSE	R
3 days	0.052	0.229	0.964
7 days	0.044	0.210	0.981
14 days	0.036	0.190	0.996
30 days	0.034	0.186	0.996

The results of the 30 -days forecast are presented using the Transformer and LSTM model in Figure 6 to show a graph of the superior prediction in 30 days. In Table III, a proportion of the performance of the Transformer model and the LSTM model is shown, including the values of MSE, RMSE, and R a period of 30 days.

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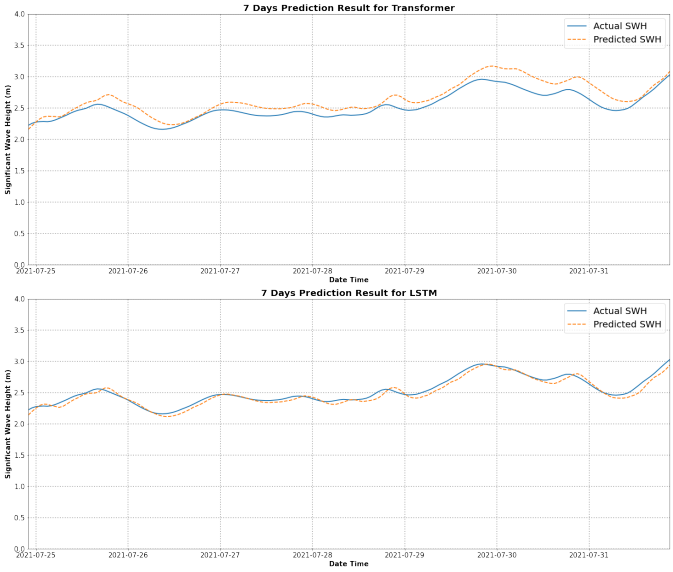


Figure 5. Results of 7 days prediction by using Transformer (upper part) and LSTM (lower part).

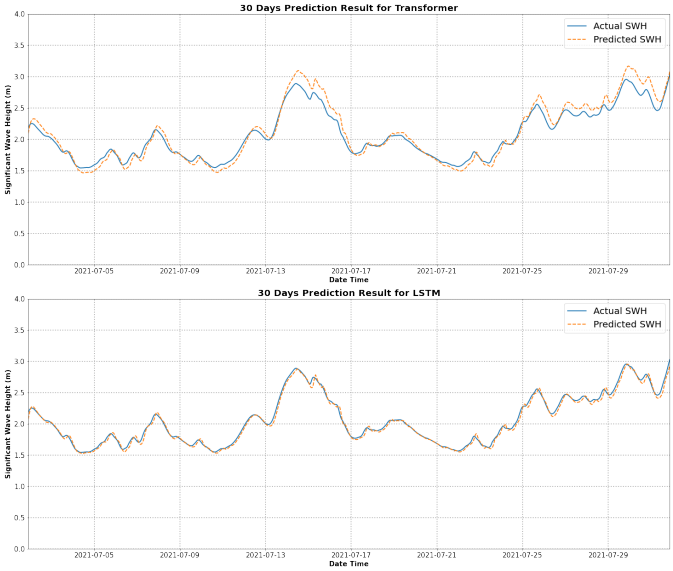


Figure 6. Results of 30 days prediction by using Transformer (upper part) and LSTM (lower part).

TABLE III
RESULTS OF PREDICTION BY USING TRANSFORMER AND LSTM MODEL TO FORECAST 30 DAYS AHEAD.

Model	MSE	RMSE	R
Transformer	0.118	0.186	0.992
LSTM	0.034	0.186	0.996

Table III shows that the LSTM model produces slightly better results compared to the Transformer model, with performance values of 0.034 for MSE, 0.186 for RMSE, and 0.996 for Correlation Coefficient.

V. CONCLUSION

This paper presents the wave height predictions as ratios using the Transformer model and the LSTM model. The wave data used in this experimentation was obtained from Pelabuhan Ratu, Indonesia, which spans from January 1, 2014 until July 31, 2021. The experimental results show that the Transformer and LSTM model have the worst and best accuracy at the same duration, 30 days for highest accuracy and three days for lowest accuracy. This is because the longer the period, the lower the accuracy. A proportion between the Transformer and LSTM model was carried out. The result is that the highest accuracy values for the Transformer model are 0.118 for MSE, 0.186 for RMSE, and 0.992 for R, while the highest values for the LSTM model are 0.034 for MSE, 0.186 for RMSE, and 0.996 for R. Transformer modification in the time series dataset has not been able to beat the performance in LSTM. Transformer model is basically better to use in the case of NLP. Based on the results of this experiment, the LSTM model is proven to be slightly superior compared to the Transformer model. Both models are relatively good at predicting wave heights for the long term forecasting.

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